# Chengyu Huang

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## Education

**Xidian University** **Xi'an, China**
*Undergraduate in Artificial Intelligence* *Sep. 2024 – Jun. 2028 (expected)*

- **GPA: 3.9/4.0** **CET-6:** 615
- **Rank: 1/283** (Computer (Intelligence) Category, 2024–2025) **1/153** (Artificial Intelligence)
- **Core Courses:** Advanced Mathematics A(I) (99), Programming Fundamentals B (99), Linear Algebra A (99), Probability Theory (98), Physics (I) (98), Data Structures (97), Discrete Mathematics (96), Advanced Mathematics A(II) (95), Introduction to Artificial Intelligence (94), Mathematical Foundations of Artificial Intelligence (92).

## Research Experience

**The Human Lab** **Tsinghua University**
*Intern, with Mingzhi Pei, Jingyan Zhang, Yao Feng.* *Aug. 2026 – Present*

- Conducting research on 4D reconstruction.

**Federated Learning** **Xidian University**
*Intern with Xu Zhang* *Sep. 2025 – Jul. 2026*

- Studied the foundations of artificial intelligence and machine learning, focusing on federated learning, personalized learning, and robust learning under label noise.

## Publication

- **Personalized Federated Learning with Noisy Labels via Total Variation Regularization**
  **Chengyu Huang**, Xu Zhang
  We propose pFedTV and cFedTV, which combine total variation regularization, noise transition modeling, and adaptive client clustering for robust personalized federated learning with heterogeneous data and noisy labels.
  *Manuscript under review, 2026.*

## Selected Projects

**PaperUnfold** *May 2026 – Jun. 2026*

- Developed the initial concept, workflow requirements, and evaluation criteria for PaperUnfold, a Codex skill for deep analysis of AI papers.
- Designed its outputs, including Chinese reading reports, condensed PDFs, and editable diagrams of paper structure and technical lineage, while separating source evidence from analysis and external verification.
  [code]

**The Listening Storyteller: An Interactive Storytelling System Powered by LLMs and Expressive Speech Synthesis** *Mar. 2026 – Jul. 2026*

- Proposed the project concept and led its design and development; built the frontend with HTML, CSS, and JavaScript and implemented the core backend functions in Python using large model APIs.
- Developed an integrated voice interaction pipeline supporting speech recognition, semantic and emotional understanding, story matching, streaming narration, voice activity detection, and spoken interruption during playback.

[code]

**AI Image Style Studio** *Jun. 2026 – Jul. 2026*

- Built a Streamlit web application with seven OpenCV artistic filters, image upload, original-result comparison, and PNG download.

[code]

**Remote Sensing Image Change Detection Using the Segment Anything Model (SAM)** *Apr. 2025 – Oct. 2025*

- Participated in a provincial undergraduate innovation project, assisting with reproduction of the SAM-SCD method and data preprocessing for remote sensing image change detection.

## Awards

- **National Scholarship** **2025**
- **Outstanding Student, Xidian University** **2025**
- **Provincial First Prize, Contemporary Undergraduate Mathematical Contest in Modeling** **2025**